

A **NOT IN** query for new customers came back empty

NOT EXISTS proved that zero was actually true

ATLIQ HARDWARE GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst



Problem Statement

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- Yesterday : Two LEFT JOIN gave an honest forecast vs actuals comparison
- Today sales wants new fy2021 customers - anyone who bought in FY2021 but not in FY2020

The Trap ❌

- NOT IN with subquery of FY2020 customers looks like fix query but it is not

Result : 0 rows returned

one NULL customer_code collapses the query



Requirment

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Check each FY2021 customers against FY2020,
one row at a time



No static list means no NULL can collapse
the comparison



NOT EXISTS, correlated and NULL safe

The Fix

AND NOT EXISTS



```
SELECT
DISTINCT fy21.customer_code
FROM fact_sales_monthly fy21
WHERE fy21.fiscal_year = 2021
AND NOT EXISTS (
SELECT 1
FROM fact_sales_monthly fy20
WHERE fy21.customer_code = fy20.customer_code
AND fy20.fiscal_year = 2020
)
```



Results

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fact_sales_monthly
dim_customer
SQL File 7*

Limit to 50000 rows

```

7
8 • SELECT
9   DISTINCT fy21.customer_code
10  FROM fact_sales_monthly fy21
11  WHERE fy21.fiscal_year = 2021
12  AND NOT EXISTS (
13    SELECT 1
14    FROM fact_sales_monthly fy20
15    WHERE fy21.customer_code = fy20.customer_code
16    AND fy20.fiscal_year = 2020
17  )
18

```

Result Grid

Filter Rows:
Export:

Wrap Cell Content:

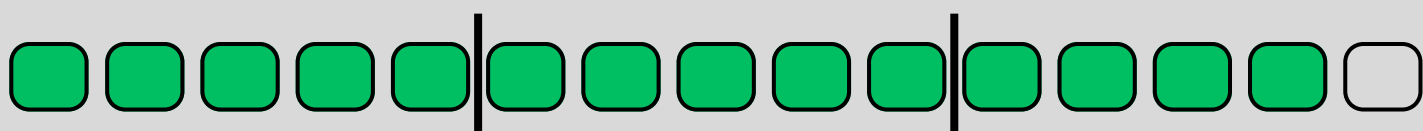
customer_code

Result 27
fact_sales_monthly 28

Output

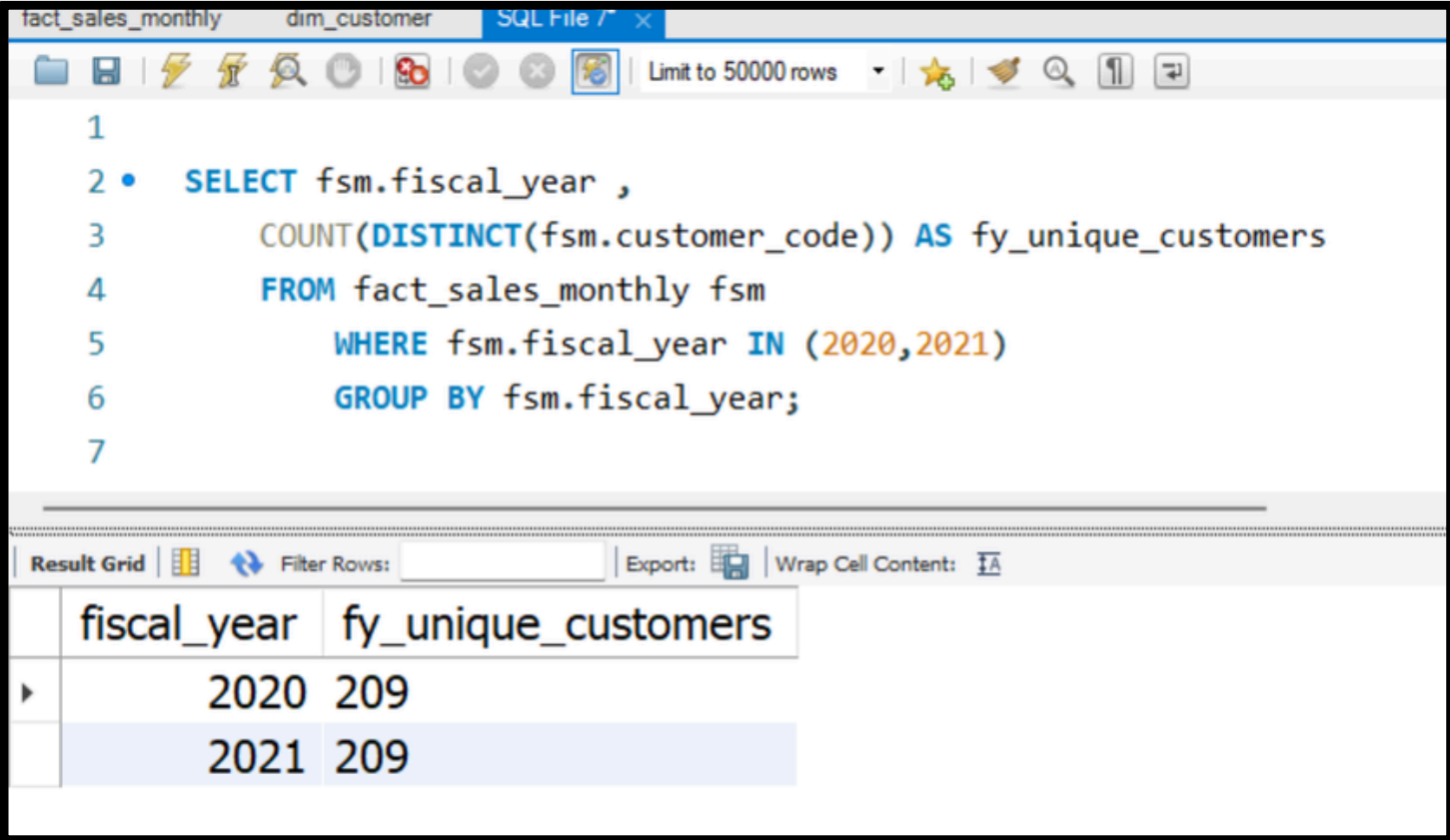
Action Output

#	Time	Action	Message
1	15:04:58	SELECT fsm.fiscal_year , COUNT(DISTINCT(fsm.customer_code)) AS fy_unique_customers FROM fact_sales_monthly ...	2 row(s) returned
2	15:05:01	SELECT DISTINCT fy21.customer_code FROM fact_sales_monthly fy21 WHERE fy21.fiscal_year = 2021 AND NOT ...	0 row(s) returned



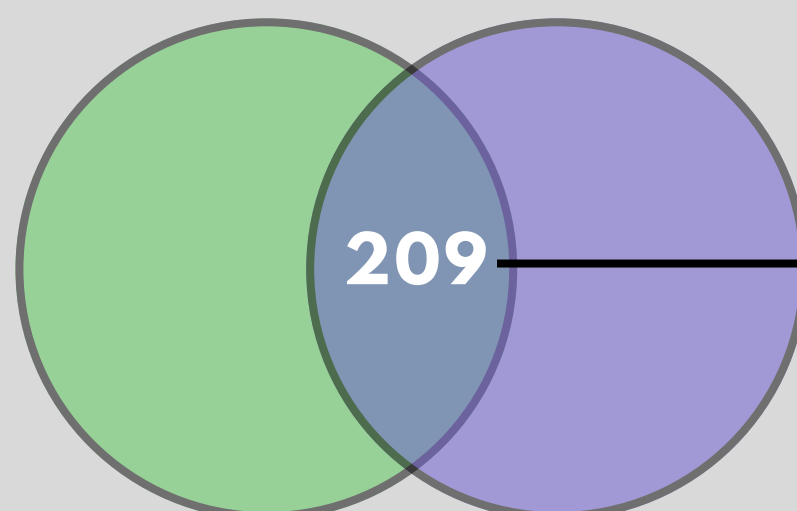
| Vishnu Ram Sachin | Data Analyst

```
SELECT fsm.fiscal_year ,  
COUNT(DISTINCT(fsm.customer_code)) AS fy_unique_customers  
FROM fact_sales_monthly fsm  
WHERE fsm.fiscal_year IN (2020,2021)  
GROUP BY fsm.fiscal_year;
```



The screenshot shows a SQL IDE window with a query editor and a results grid. The query is the same as the one in the previous block. The results grid shows two rows of data.

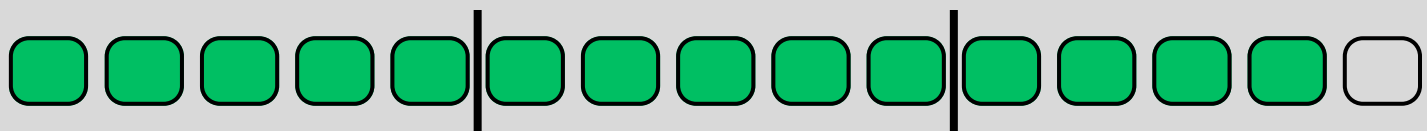
fiscal_year	fy_unique_customers
2020	209
2021	209

FY2020**FY2021****209****Same customers****So confirmed 0 new customers**



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SQL THROUGH ONE DATASET CHALLENGE